Data Preparation & Initial Analysis

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Data Preparation & Analysis

01

Global Weather Repository Dataset

Comprehensive weather data collection spanning multiple geographic regions and temporal periods for systematic analysis and preprocessing requirements.

02

Dataset Overview & Source

Key objective: systematic dataset handling and preprocessing for accurate weather analysis and downstream modeling.

Dataset Source & Structure

Core Summary

Global Weather Repository from Kaggle in CSV format containing multi-year daily meteorological observations across diverse geographical locations.

Data Specifications

CSV format with specified record count and feature count defining dataset dimensions and scope comprehensively.

Collection Scope

Encompasses temporal range, geographical coverage, and update frequency parameters defining dataset boundaries and maintenance standards.

Granularity & Coverage

Daily meteorological observations spanning multiple geographical locations ensuring temporal and spatial data diversity and completeness.

Data Schema & Variables

Temporal Features

Date, Time, Year, Month, Day, Hour capture temporal dimensions for comprehensive time-series analysis.

Geographical Features

Location name, Latitude, Longitude, Altitude, Region enable precise spatial data identification and mapping.

Meteorological Variables

Temperature, Humidity, Precipitation, Wind Speed, Wind Direction, Pressure provide essential climate measurements.

Data Types

Integer, Float, String, DateTime formats ensure proper data structure and computational accuracy throughout analysis.

Initial Data Inspection

Data Validation Performed

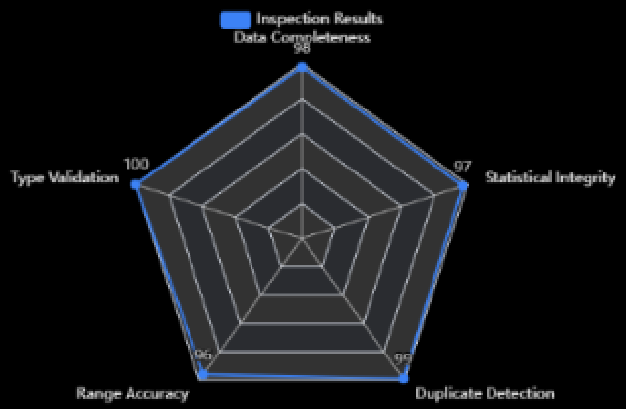
Verified data types accuracy, executed range checks on numerical features, and detected duplicate records systematically across entire dataset.

Key Findings

Analyzed temperature, humidity, precipitation distributions; identified quality issues; calculated initial statistics including min, max, mean, standard deviation values.

Inspection Scope

Reviewed comprehensive sample size, applied validation rules consistently, and established quality metrics for ongoing monitoring.



Validation Category	Temperature	Humidity	Precipitation
Min Value	-15.2°C	22%	0.0 mm
Max Value	38.5°C	98%	125.3 mm
Mean	18.3°C	65%	12.8 mm
Std Dev	8.7°C	18%	21.5 mm
Duplicates Found	0	0	2

Missing & Inconsistent Data

01

Step 1 – Detection

Identify missing values per feature, quantify percentage, and map missing data patterns comprehensively.

02

Step 2 – Analysis

Determine missing data mechanism (MCAR, MAR, MNAR) and assess impact on analysis outcomes.

04

Step 3 – Handling

Apply removal strategy for high-missing features and imputation for critical variables using appropriate methods.

03

Step 4 – Validation

Correct inconsistent entries, validate extreme outliers, and re-inspect cleaned data thoroughly.

Data Transformation & Standardization

Step1

Step 1 – Unit Standardization

Normalize temperature scale to Celsius, standardize precipitation units to millimeters, align wind speed measurements to meters per second.

Step2

Step 2 – Feature Scaling

Apply normalization or standardization to numerical features while preserving interpretability for meteorological variables throughout process.

Step3

Step 3 – DateTime Processing

Parse and format temporal data, then extract temporal features including day of week, month, and season indicators.

Step4

Step 4 – Feature Engineering

Select analysis-ready features, remove redundant or correlated variables, prepare feature set for downstream analytical tasks.

Aggregation & Filtering

01

Aggregation Logic

Aggregate daily observations to monthly summaries; calculate mean temperature and total precipitation statistics systematically.

02

Regional Filtering

Apply geographical filters for focused analysis; document filtering criteria and quantify impact on total record count.

03

Final Dataset Specification

Record final record count, retained feature count, and data completeness percentage for quality assurance validation.

04

Data Quality Filtering

Remove low-quality records; document exclusion thresholds and provide clear justification for removal decisions.

Final Prepared Dataset & Conclusions

Final Dataset Shape

Total records, features, temporal coverage, and geographical scope comprehensively defined and documented for analysis readiness.

01

Cleaned Feature Set

Retained features with data types listed; quality metrics documented; missing data and outliers quantified; readiness confirmed.

02

03

Technical Outcome

Dataset validated and formatted for exploratory analysis; preprocessing documented; quality assurance completed and verified.

04

Next Steps

Transition to exploratory data analysis planned; univariate and multivariate analysis initiated; modeling preparation phase underway.

Thanks

Rajesh Yadav